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Estimation of Dynamic Models With Error Components

T. W. ANDERSON and CHENG HSIAO*

Observations on N cross-section units at T time points are used to estimate a simple statistical model involving an autoregressive process with an additive term specific to the unit. Different assumptions about the initial conditions are (a) initial state fixed, (b) initial state random, (c) the unobserved individual effect independent of the unobserved dynamic process with the initial value fixed, and (d) the unobserved individual effect independent of the unobserved dynamic process with initial value random. Asymptotic properties of the maximum likelihood and "covariance" estimators are obtained when $T \rightarrow \infty$ and when $N \rightarrow \infty$. The relationship between the pseudo and conditional maximum likelihood estimators is clarified. A simple consistent estimator that is independent of the initial conditions and the way in which T or $N \rightarrow \infty$ is also suggested.

KEY WORDS: Dynamic models; Error components; Cross-section time series; Initial conditions.

1. INTRODUCTION

Error components models are widely used in the econometric analysis of cross-section and time series data; it is often assumed that the large number of factors that affect the individuals in the sample and the values of the dependent variable observed for them, but that have not been explicitly included as independent variables, may be appropriately summarized by random disturbances.¹ Wallace and Hussain (1969) and Swamy and Arora (1972) have analyzed this type of model when no lagged dependent variables appear as explanatory variables. Tiao and Ali (1971) have also studied a related model.

Very often we would like to use such a model to study

behavioral relationships that are dynamic in character (Balestra and Nerlove 1966). As it turns out, the problem becomes complicated. Many people have studied various aspects of the model. In this paper more attention is paid to the properties of various estimators according to the way in which the time series observations T or the cross-sectional units N tend to infinity. In the typical case in which a panel involves a large number of individuals, but only over a short period of time, the relevant limiting distributions should have the number of individuals increasing but not the time. Hence, contrary to the dynamic model for a single time series, the assumption about the initial observations plays a crucial role in interpreting the model and devising consistent estimates. In this paper we shall consider a number of different models arising from different assumptions about the initial observations. The focus will be on (a) the interpretation of the model and (b) the asymptotic properties of the estimators under various assumptions.

2. THE MODEL

A simple model commonly used in the empirical research of a sample consisting of time series observations on a cross-section is of the form²

$$y_{it} = \mathbf{z}_{it}\boldsymbol{\beta} + v_{it}, \quad i = 1, \dots, N, \\ t = 1, \dots, T, \quad (2.1)$$

where $\mathbf{z}_{it}$ is an $1 \times k$ vector of explanatory variables; $\boldsymbol{\beta}$ is a $k \times 1$ vector of parameters; $v_{it} = \alpha_i + u_{it}$; $E\alpha_i = Eu_{it} = 0$; $E\alpha_i u_{jt} = 0$; $E(\alpha_i \alpha_j) = \sigma^2$ if $i = j$, and $= 0$ otherwise; and $E(u_{it} u_{js}) = \lambda \sigma^2 = \sigma_u^2$ if $i = j$, $t = s$, and $= 0$ otherwise. We are interested in cases in which $T \geq 2$, $N \geq 2$. Let $\mathbf{y}_i = (y_{i1}, \dots, y_{iT})'$, $\mathbf{Z}_i = (\mathbf{z}'_{i1}, \mathbf{z}'_{i2}, \dots, \mathbf{z}'_{iT})'$, $\mathbf{v}_i = (v_{i1}, \dots, v_{iT})'$, $\mathbf{u}_i = (u_{i1}, \dots, u_{iT})'$, and $\mathbf{e} = (1, \dots, 1)'$. We can rewrite (2.1) as

$$\mathbf{y}_i = \mathbf{Z}_i \boldsymbol{\beta} + \mathbf{v}_i, \quad i = 1, \dots, N. \quad (2.2)$$

Premultiplying (2.2) by $\mathbf{Q} = \mathbf{I}_T - (1/T)\mathbf{e}\mathbf{e}'$, we obtain

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¹ The attempt to account for the heterogeneity among the cross-sectional units by error components model does not always yield economically meaningful results. For example, see Mehta, Narasimham, and Swamy (1978).

² We assume no correlation between the unobserved effects and the observed explanatory variables; this assumption is unlike that of Mundlak (1978). As will be discussed later, we essentially follow a different interpretation of the model from that of Mundlak.

the covariance estimator (CV) of β as

$$\hat{\beta}_{CV} = \left[\sum_{i=1}^N \mathbf{Z}'_i \mathbf{Q} \mathbf{Z}_i \right]^{-1} \sum_{i=1}^N \mathbf{Z}'_i \mathbf{Q} \mathbf{y}_i. \quad (2.3)$$

If we assume that α_i and u_{it} are normally distributed, we can write down the likelihood function of $\mathbf{y}_i$ from the density function of $\mathbf{v}_i$. Maximizing the logarithm of the likelihood function, we obtain the maximum likelihood estimator (MLE) of β , σ^2 , and λ .

For $\mathbf{Z}_i$ exogenous, Wallace and Hussain (1969) proved that the CV is consistent and asymptotically normally distributed when $N \rightarrow \infty$ or $T \rightarrow \infty$ or both. Furthermore, the CV is asymptotically equivalent to the MLE as long as $T \rightarrow \infty$ (and N is fixed or tends to infinity).

When $\mathbf{z}_{it}$ contains lagged endogenous variables, the problem becomes more complicated. Not only may the CV and the MLE be inconsistent, but the interpretation of the model is dependent on the assumption about the initial conditions. In this paper we shall assume that $\mathbf{z}_{it}$ consists of $y_{i,t-1}$ only (namely, $k = 1$) because the principle of analysis remains the same, yet the presentation can be greatly simplified. Therefore, instead of (2.1) we shall analyze

$$y_{it} = \beta y_{i,t-1} + \alpha_i + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \quad (2.4)$$

We also assume that $|\beta| < 1$ and that the mean of the observed variable is known (and taken equal to zero).³

We have essentially four different cases based on different assumptions about the initial observations y_{i0} in (2.4). The first case is a conventional one in which we assume that y_{i0} are observed fixed constants (Amemiya 1967; Balestra and Nerlove 1966, etc.). This assumption permits a cross-sectional unit starting at some y_{i0} and gradually moving toward a level of $[\alpha_i/(1 - \beta)]$. To see this, we rewrite (2.4) in an equivalent form of

$$(y_{it} - \gamma_i) = \beta(y_{i,t-1} - \gamma_i) + u_{it}, \quad (2.5)$$

where

$$\alpha_i = (1 - \beta)\gamma_i, \quad E\gamma_i = 0, \quad \text{var}(\gamma_i) = \sigma_{\gamma}^2 = \sigma^2/(1 - \beta)^2. \quad (2.6)$$

Then the assumption of fixed initial conditions implies that a cross-sectional unit may start from an arbitrary initial position, and gradually move towards its respective level γ_i according to a probability law. The individual level γ_i is a random draw from a population with mean zero and variance σ_{γ}^2 . This is a reasonable model, but there might be a question of treating y_{i0} as fixed if the decision when to start sampling is arbitrary and, in particular, independent of the value of y_{i0} .

An alternative formulation is to assume that the initial observations are random with a common mean (Case II). For instance, we may assume that

$$y_{i0} = c + \epsilon_i, \quad i = 1, \dots, N. \quad (2.7)$$

Then we may say that ϵ_i represents the effect of initial individual endowments (corrected for the mean). Depending on the assumption about ϵ_i , the impact of initial endowments will be different as time goes by. If we assume ϵ_i to be random with mean zero and variance σ_{ϵ}^2 , and to be independent of α_i and u_{it} , its impact gradually diminishes and eventually vanishes. The model is somewhat like Case I, in which the starting value and the level γ_i are independent, except that now the starting observable value is not a fixed constant but a random draw from a population with mean c and variance σ_{ϵ}^2 . On the other hand, if we wish to allow the initial endowment affecting the level, we may let

$$y_{i0} = c + \alpha_i, \quad i = 1, \dots, N. \quad (2.8)$$

Then, as time goes by, the effect of initial endowments cumulates and eventually reaches a level of $[\alpha_i/(1 - \beta)]$

A different formulation allowing dependence between α_i and y_{i0} is to rewrite (2.4) as

$$y_{it} = w_{it} + \gamma_i, \quad t = 0, 1, \dots, T, \quad (2.9)$$

$$w_{it} = \beta w_{i,t-1} + u_{it}, \quad t = 1, \dots, T, \quad (2.10)$$

where w_{it} , γ_i , and u_{it} are unobservable. It is convenient in (2.4) to assume that the starting observable value and the level are independent. In the form (2.9) and (2.10) it is convenient to assume that level γ_i and the unobservable process $\{w_{it}\}$ are independent; then the starting value y_{i0} is correlated with the level γ_i . If we allow correlation between y_{i0} and γ_i , the two models are equivalent by (2.7) and (2.8).

Analogous to the assumption that y_{i0} may be fixed or random is the assumption that w_{i0} are either fixed or random. When w_{i0} are assumed fixed (Case III), it is similar to Case I, in which an individual may start at some value y_{i0} and move toward a level of γ_i , except that in Case III the individual equilibrium level γ_i affects the initial value.

When w_{i0} are assumed random (Case IV), alternative standard assumptions about w_{i0} are (a) stationary with variance $\lambda\sigma^2/(1 - \beta^2)$ and (b) random with arbitrary variance $(\lambda/\eta)\sigma^2$. In subcase (a), y_{i0} is considered to have the marginal normal distribution determined by the stationary process; that is, y_{i0} is viewed to form part of a stationary process as any other y_{it} .

3. FIXED INITIAL OBSERVATIONS

In this section we assume that the initial observations y_{i0} are fixed constants and observable such that $\lim_{N \rightarrow \infty} \sum_{i=1}^N y_{i0}^2/N$ exists. Then $\text{plim}_{N \rightarrow \infty} \sum_{i=1}^N [\sum_{t=1}^T y_{i,t-1}^2]$

³ The stationarity assumption may be relaxed when T is fixed and $N \rightarrow \infty$ (e.g., see Anderson 1978). We keep this assumption for simplicity of exposition and because it allows us to provide a unified approach toward various assumptions about the initial conditions discussed later.

$-T\bar{y}_{i,-1}^2]/NT$ exists and is nonzero. We define $\bar{y}_i = \sum_{t=1}^T y_{it}/T$, $\bar{y}_{i,-1} = \sum_{t=1}^T y_{i,t-1}/T$.

We first consider the property of the CV of β . The CV for (2.4) is obtained by solving the following normal equation:

$$\sum_{i=1}^N \left[\frac{1}{T} \sum_{t=1}^T y_{i,t-1}^2 - \bar{y}_{i,-1}^2 \right] \hat{\beta}_{CV} = \sum_{i=1}^N \left[\frac{1}{T} \sum_{t=1}^T y_{it}y_{i,t-1} - \bar{y}_i\bar{y}_{i,-1} \right]. \quad (3.1)$$

We note that $\mathbf{Qv}_i = \mathbf{Qu}_i$. The solution of (3.1) is the least squares estimator of β of (2.4) with the interpretation that α_i are unknown constants. Thus when $T \rightarrow \infty$ (regardless of whether N is fixed or tends to infinity), we can prove the consistency and asymptotic normality of the covariance estimator in exactly the same form as that of Anderson (1971, Ch. 5, Sec. 5.5, pp. 200–203). The variance of the limiting distribution of $\sqrt{NT}(\hat{\beta}_{CV} - \beta)$ is

$$\lambda\sigma^2 \left\{ \text{plim}_{T \rightarrow \infty} \frac{1}{NT} \sum_{i=1}^N \left[\sum_{t=1}^T y_{it}^2 - T\bar{y}_i^2 \right] \right\}^{-1} = 1 - \beta^2. \quad (3.2)$$

On the other hand, when T is fixed but $N \rightarrow \infty$, the CV is inconsistent. This can be seen by noting that

$$\hat{\beta}_{CV} = \beta + \frac{\frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T y_{i,t-1}u_{it} - \frac{1}{N} \sum_{i=1}^N \bar{y}_{i,-1}\bar{u}_i}{\frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T y_{i,t-1}^2 - \frac{1}{N} \sum_{i=1}^N \bar{y}_{i,-1}^2}, \quad (3.3)$$

where $\bar{u}_i = \sum_{t=1}^T u_{it}/T$. By a law of large numbers

$$\text{plim}_{N \rightarrow \infty} \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T y_{i,t-1}u_{it} = 0, \quad (3.4)$$

$$\begin{aligned} \text{plim}_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \bar{y}_{i,-1}\bar{u}_i \\ = \frac{\lambda\sigma^2 T - 1 - T\beta + \beta^T}{T^2(1 - \beta^2)} \neq 0. \end{aligned} \quad (3.5)$$

The CV of β is also the MLE under the assumption that α_i are fixed constants and u_{it} are normally distributed. (Then the mean of the observed variable is not known and is not taken equal to zero.) In this paper, however, we assume that α_i are random. When α_i and u_{it} are normally distributed, we can write the logarithm of the likelihood function as

$$\begin{aligned} \log L = -\frac{NT}{2} \log 2\pi - \frac{NT}{2} \log \sigma^2 - \frac{N}{2} \log |\Lambda| \\ - \frac{1}{2\sigma^2} \sum_{i=1}^N (\mathbf{y}_i - \mathbf{y}_{i,-1}\beta)' \Lambda^{-1} (\mathbf{y}_i - \mathbf{y}_{i,-1}\beta), \end{aligned} \quad (3.6)$$

where

$$\mathbf{y}_{i,-1} = (y_{i0}, y_{i1}, \dots, y_{i,T-1})',$$

$$\Lambda_{T \times T} = \lambda \mathbf{I}_T + \mathbf{ee}', \quad (3.7)$$

$$\Lambda^{-1} = \frac{1}{\lambda} \mathbf{I}_T - \frac{1}{\lambda(T + \lambda)} \mathbf{ee}'$$

$$|\Lambda| = \lambda^{T-1}(T + \lambda),$$

and $\mathbf{I}_T$ is the $T \times T$ identity matrix.

The MLE is obtained by solving the first-order conditions of (3.6) with respect to β , σ^2 , and λ .⁴ If there is a unique solution to these partial derivative equations with $\sigma^2 > 0$ and $0 < \lambda < \infty$, the solution is the MLE. Maddala (1971) showed that the boundary solution of $\lambda = 0$ (that is, $\sigma_u^2 = 0$) cannot occur, but the boundary solution of $\sigma^2 = 0$ (that is, $\lambda = \infty$) will occur when $\sum_{i=1}^N \sum_{t=1}^T (y_{it} - \beta y_{i,t-1})^2 \geq T^2 \sum_{i=1}^N (\bar{y}_i - \beta \bar{y}_{i,-1})^2$. The solution, then, does not satisfy all the derivative equations. Nerlove (1971) obtained such solutions in a simulation study. However, the probability of a boundary solution tends to zero as $T \rightarrow \infty$ or as $N \rightarrow \infty$.⁵

Amemiya (1967) and Balestra and Nerlove (1966) showed that the MLE of β is consistent and asymptotically normally distributed when $T \rightarrow \infty$ (and N is fixed or tends to infinity). When T is fixed and $N \rightarrow \infty$, it can be shown that the MLE is consistent and asymptotically normally distributed under the assumption that y_{i0} are fixed constants.

To show that the MLE is consistent when N tends to infinity and T is fixed, we let $\theta = (\beta, \sigma^2, \lambda)'$. Noting that $E(u_{it}y_{i,t+j}) = \beta^j \lambda \sigma^2$ for $j \geq 0$, and $E(\alpha_i y_{it}) = (1 + \beta + \dots + \beta^{t-1})\sigma^2$ for $t \geq 2$ and $E\alpha_i y_{i1} = \sigma^2$, we can show that $[1/(NT)](\partial \log L / \partial \theta)$ converges in probability to $\mathbf{0}$ at the true θ when T is fixed and $N \rightarrow \infty$. Furthermore, $[1/(NT)](\partial^2 \log L / \partial \theta \partial \theta')$ around its true value converges in probability to (-1) times

$$\begin{pmatrix} \frac{1}{\lambda\sigma^2} \text{plim}_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{T} \sum_{t=1}^T y_{i,t-1}^2 - \frac{T}{\lambda + T} \bar{y}_{i,-1}^2 \right] & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{2\sigma^4} & \frac{\lambda + (T-1)}{2\sigma^2\lambda(\lambda + T)} & \frac{\lambda + (T-1)}{2\sigma^2\lambda(\lambda + T)} \\ \frac{\lambda + (T-1)}{2\sigma^2\lambda(\lambda + T)} & \frac{[\lambda^2 + 2\lambda(T-1) + T(T-1)]}{2\lambda^2(\lambda + T)^2} & \frac{[\lambda^2 + 2\lambda(T-1) + T(T-1)]}{2\lambda^2(\lambda + T)^2} \end{pmatrix}, \quad (3.8)$$

⁴ For this and many other technical details of results mentioned in this paper, see Anderson and Hsiao (1980).

⁵ An alternative parameterization that allows the correlation between α_i and u_{it} would avoid this question of boundary solution, but then only β and σ_u^2 are identified. The parameters $E(\alpha_i u_{it})$ and σ^2 are not identified, although the sum $\sigma^2 + 2E(\alpha_i u_{it})$ is identifiable. See Berzecz (1979), Swamy (1974), and Swamy and Mehta (1973), for example.

which is negative definite as long as $T \geq 2$. Thus, there is a consistent root for the MLE (Foutz 1977). We can also show the asymptotic normality of the MLE of β by an argument similar to that of Anderson (1978) or Miller (1977).

The solution for the MLE is complicated. An iterative procedure such as the Newton-Raphson type will have to be used. However, if $T \rightarrow \infty$, the CV is not only consistent but is also asymptotically equivalent to the MLE. (See (3.2) and 3.8).)

It is interesting to note that when N is fixed and $T \rightarrow \infty$, it is not possible to obtain separate consistent estimates of σ^2 and λ (Amemiya 1967). Yet if T is fixed, as long as it is greater than one, we can get separate consistent estimates of σ^2 and λ when $N \rightarrow \infty$. When $T = 1$, we have N independent random variables $v_{i1} = \alpha_i + u_{i1}$ with variance $\sigma^2(1 + \lambda)$. It is not possible to distinguish λ and σ^2 . However, the MLE of β in this case is consistent and it is the same as the least squares estimate.

The incidental parameters problem does not arise under the assumption that y_{i0} 's are fixed because they are observed. The individual component α_i only gives rise to a special form of the covariance matrix of $\mathbf{v}_i$. The consistency of the MLE is a consequence of the fact that we are maximizing the likelihood function of N random vectors $\mathbf{y}_i$ that are independently normally distributed.

4. RANDOM INITIAL OBSERVATIONS WITH A COMMON MEAN

In this section we first consider the model (2.7) where the initial endowments ϵ_i do not affect the level γ_i and the disturbances u_{it} . Then the joint likelihood function of $(y_{i0}, \dots, y_{iT})$ is

$$\begin{aligned} f_i(y_{i0}, \dots, y_{iT}) &= f_i(y_{i1}, \dots, y_{iT} | y_{i0}) f_i(y_{i0}) \\ &= (2\pi\sigma^2)^{-T/2} | \Lambda |^{-1/2} \exp \left\{ -(1/2\sigma^2) \right. \\ &\quad \times (\mathbf{y}_i - \mathbf{y}_{i,-1}\beta)' \Lambda^{-1} (\mathbf{y}_i - \mathbf{y}_{i,-1}\beta) \left. \right\} \\ &\quad \times (2\pi\sigma_\epsilon^2)^{-1/2} \exp \left\{ -(1/2\sigma_\epsilon^2) (y_{i0} - c)^2 \right\}. \end{aligned} \quad (4.1)$$

Therefore, the MLE of β , λ , and σ^2 is identical to the MLE under the assumption that y_{i0} are fixed constants; however, in addition to estimating these parameters, we also estimate c and σ_ϵ^2 by

$$\hat{c} = \sum_{i=1}^N y_{i0}/N, \quad \hat{\sigma}_\epsilon^2 = \sum_{i=1}^N (y_{i0} - \hat{c})^2/N. \quad (4.2)$$

On the other hand, if we assume (2.8), the joint likelihood function is

$$\begin{aligned} f_i(y_{i0}, \dots, y_{iT}) &= f_i(y_{i1}, \dots, y_{iT} | y_{i0}) f_i(y_{i0}) \\ &= (2\pi)^{-T/2} (\sigma^2 \lambda)^{-T/2} \end{aligned} \quad (4.3)$$

$$\begin{aligned} &\exp \left\{ -\frac{1}{2\lambda\sigma^2} \sum_{t=1}^T [(y_{it} - y_{i0}) \right. \\ &\quad \left. - \beta y_{i,t-1} + c]^2 \right\} \cdot (2\pi)^{-1/2} (\sigma^2)^{-1/2} \\ &\exp \left\{ -\frac{(y_{i0} - c)^2}{2\sigma^2} \right\}. \end{aligned}$$

The MLE of $\delta = (\beta, c, \sigma^2, \lambda)'$ is obtained by taking the partial derivatives of the logarithm of $\prod_{i=1}^N f_i(y_{i0}, \dots, y_{iT})$ and setting them equal to zero. Contrary to Case I, the solution to the derivative equations is always an interior one, and there is no boundary value problem.

It is easy to show that $[1/(N(T+1))] \cdot [(\partial \log L)/\partial \delta]$ converges in probability to $\mathbf{0}$ at the true value and $[1/(N(T+1))] \cdot [(\partial^2 \log L)/\partial \delta \partial \delta']$ converges in probability to a negative definite matrix when either $T \rightarrow \infty$ and N is fixed or $N \rightarrow \infty$ and T is fixed or both. Therefore, the MLE is consistent in either case.

We note that conditional on y_{i0} in (4.3) we can maximize $L^* = \prod_{i=1}^N f_i(y_{i1}, \dots, y_{iT} | y_{i0})$ with respect to β , c , and $\lambda\sigma^2$. This conditional MLE is consistent when either T or N or both tend to infinity and is asymptotically normally distributed. Of course, when T is fixed and $N \rightarrow \infty$, the unconditional MLE of β and c is more efficient than the conditional MLE (in the sense of having smaller variance-covariance matrix). But it is computationally more tedious than the conditional MLE. The conditional MLE is obtained by setting the partial derivatives of $\log L^*$ equal to zero. They are equivalent to estimators based on the least squares regression of $(y_{it} - y_{i0})$ on $y_{i,t-1}$ and a constant term.⁶ This solution can either be used as a consistent estimate or be used to start the iterative procedure to obtain unconditional MLE.

The property of the CV is the same as in Case I. It is consistent when $T \rightarrow \infty$ and inconsistent when T is fixed and $N \rightarrow \infty$.

5. RANDOM INITIAL OBSERVATIONS WITH DIFFERENT MEANS

In this and the next section we shall consider models (2.9) and (2.10). We first consider the case that w_{i0} are unknown constants. Then at $t = 0$, $\gamma_i = y_{i0} - w_{i0}$. The density of $(y_{i0}, y_{i1}, \dots, y_{iT})$ is⁷

$$\begin{aligned} f_i(y_{i0}, y_{i1}, \dots, y_{iT}) &= f_i(y_{i1}, \dots, y_{iT} | y_{i0}) f_i(y_{i0}) \\ &= \left(\frac{1}{\sqrt{2\pi}\sigma_u} \right)^T \exp \left\{ -\frac{1}{2\sigma_u^2} \sum_{t=1}^T [(y_{it} - y_{i0} + w_{i0}) \right. \end{aligned}$$

⁶ Note that if the original model contains an intercept term, the conditional MLE can only provide a consistent estimate of c subtracting the intercept. Neither can the conditional MLE distinguish λ and σ^2 . The unconditional MLE can distinguish the intercept, c , λ , and σ^2 .

⁷ Note that we use σ_u^2 in place of $\lambda\sigma^2$ and σ_γ^2 in place of $\sigma^2/(1 - \beta)^2$ in this section for ease of exposition.

$$- \beta(y_{i,t-1} - y_{i0} + w_{i0})]^2 \} \quad (5.1)$$

$$\cdot \frac{1}{\sqrt{2\pi}\sigma_\gamma} \exp \left\{ -\frac{1}{2\sigma_\gamma^2} (y_{i0} - w_{i0})^2 \right\}.$$

For given y_{it} 's it is a function of w_{i0} 's, β , σ_u^2 , and σ_γ^2 . If we let $w_{i0} = y_{i0}$, then the second factor is $1/(\sqrt{2\pi}\sigma_\gamma)$. As $\sigma_\gamma \rightarrow 0$, this approaches ∞ and the likelihood function approaches ∞ . The likelihood function does not have a maximum, and hence the MLE does not exist.

However, if we focus our attention on the interior solutions, we may take the partial derivatives of the logarithm of the joint likelihood function $\prod_{i=1}^N f_i(y_{i0}, y_{i1}, \dots, y_{iT})$,

$$\begin{aligned} \log L = & -\frac{NT}{2} \log 2\pi - \frac{NT}{2} \log \sigma_u^2 \\ & - \frac{1}{2\sigma_u^2} \sum_{i=1}^N \sum_{t=1}^T [(y_{it} - y_{i0} + w_{i0}) \\ & - \beta(y_{i,t-1} - y_{i0} + w_{i0})]^2 - \frac{N}{2} \log 2\pi \\ & - \frac{N}{2} \log \sigma_\gamma^2 - \frac{1}{2\sigma_\gamma^2} \sum_{i=1}^N (y_{i0} - w_{i0})^2, \end{aligned} \quad (5.2)$$

with respect to w_{i0} , β , σ_u^2 , and σ_γ^2 . Setting these equal to zero, we obtain

$$\begin{aligned} [T(1 - \hat{\beta})^2 \hat{\sigma}_\gamma^2 + \hat{\sigma}_u^2] \hat{w}_{i0} \\ = \hat{\sigma}_u^2 y_{i0} - \hat{\sigma}_\gamma^2 (1 - \hat{\beta}) \sum_{t=1}^T [(y_{it} - y_{i0}) \\ - \hat{\beta}(y_{i,t-1} - y_{i0})], \quad i = 1, \dots, N, \end{aligned} \quad (5.3)$$

$$\begin{aligned} \sum_{i=1}^N \sum_{t=1}^T (y_{it} - y_{i0} + \hat{w}_{i0})(y_{i,t-1} - y_{i0} + \hat{w}_{i0}) \\ = \hat{\beta} \sum_{i=1}^N \sum_{t=1}^T (y_{i,t-1} - y_{i0} + \hat{w}_{i0})^2, \end{aligned} \quad (5.4)$$

$$\hat{\sigma}_u^2 = \sum_{i=1}^N \sum_{t=1}^T [(y_{it} - y_{i0} + \hat{w}_{i0}) \quad (5.5)$$

$$- \hat{\beta}(y_{i,t-1} - y_{i0} + \hat{w}_{i0})]^2 / (NT), \quad (5.6)$$

$$\hat{\sigma}_\gamma^2 = \sum_{i=1}^N (y_{i0} - \hat{w}_{i0})^2 / N.$$

We want to show first that the interior solution is asymptotically equivalent to the CV as $T \rightarrow \infty$. The coefficient of $\hat{\beta}$ on the right-hand side of (5.4) divided by NT can be written

$$\begin{aligned} \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T [(y_{i,t-1} - \bar{y}_{i,-1}) \\ + (\bar{y}_{i,-1} - y_{i0} + \hat{w}_{i0})]^2 \end{aligned} \quad (5.7)$$

$$\begin{aligned} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{T} \sum_{t=1}^T (y_{i,t-1} - \bar{y}_{i,-1})^2 \right. \\ \left. + (\bar{y}_{i,-1} - y_{i0} + \hat{w}_{i0})^2 \right]. \end{aligned}$$

From (5.3) we find

$$\hat{w}_{i0} - y_{i0} = - \frac{\hat{\sigma}_\gamma^2 (1 - \hat{\beta}) T}{T(1 - \hat{\beta})^2 \hat{\sigma}_\gamma^2 + \hat{\sigma}_u^2} (\bar{y}_i - \hat{\beta} \bar{y}_{i,-1}). \quad (5.8)$$

Then we see that the second term on the right-hand side of (5.7) converges in probability to zero. Similarly, the left-hand side of (5.4) divided by NT can be written

$$\begin{aligned} \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{T} \sum_{t=1}^T (y_{it} - \bar{y}_i)(y_{i,t-1} - \bar{y}_{i,-1}) \right. \\ \left. + (\bar{y}_i - y_{i0} + \hat{w}_{i0})(\bar{y}_{i,-1} - y_{i0} + \hat{w}_{i0}) \right], \end{aligned} \quad (5.9)$$

and the second term converges in probability to zero. Then as $T \rightarrow \infty$, (5.4) is equivalent to (3.1). The solution is consistent as $T \rightarrow \infty$.

Now let us consider the case of $N \rightarrow \infty$. If we substitute (5.8) into (5.4), (5.5), and (5.6), we obtain three polynomial equations in $\hat{\beta}$, $\hat{\sigma}_u^2$, and $\hat{\sigma}_\gamma^2$. There may be several different solutions. These give stationary points of the likelihood function; perhaps some are relative maxima. But none can give an absolute maximum. Neither does any of these relative maxima yield a consistent root. To show this, we assume that $(1/N) \sum_{i=1}^N w_{i0}^2$ converges to a finite constant. We solve (5.3) and put it into (5.4). Then we subtract the right-hand side from the left-hand side. If $\hat{\beta}$ is consistent, we can replace it by β and find the probability limit of this difference divided by NT . Its probability limit is not equal to zero; it is equal to

$$\begin{aligned} \frac{-\sigma_u^2 \hat{\sigma}_\gamma^2 [T - 1 - T\beta + \beta^T]}{[T(1 - \beta)^2 \hat{\sigma}_\gamma^2 + \hat{\sigma}_u^2] T} \\ + \frac{(1 - \beta) \hat{\sigma}_u^2 [\hat{\sigma}_u^2 \sigma_\gamma^2 - \sigma_u^2 \hat{\sigma}_\gamma^2]}{[T(1 - \beta)^2 \hat{\sigma}_\gamma^2 + \hat{\sigma}_u^2]^2}. \end{aligned} \quad (5.10)$$

This contradiction shows that $\hat{\beta}$ is not consistent.

The analysis of the behavior of the CV proceeds as in the two previous sections. The CV is consistent as $T \rightarrow \infty$ and is inconsistent as $N \rightarrow \infty$.

It may be of interest to note that when $T = 1$, this is similar to the classical problem of incidental parameters.

6. RANDOM INITIAL OBSERVATIONS WITH A STATIONARY DISTRIBUTION

In this section we consider (2.9) and (2.10) with w_{i0} random. We first consider subcase (a) where w_{i0} is assumed to be normally distributed with mean zero and variance $\lambda\sigma^2/(1 - \beta^2)$. Then y_{i0} will be normally distributed with mean zero and variance $\lambda\sigma^2/(1 - \beta^2) + \sigma^2/(1 - \beta)^2$, and $E(\alpha_i y_{i0}) = \sigma^2/(1 - \beta)$. The joint density of

$(y_{i0}, y_{i1}, \dots, y_{iT})$ is

$$f_i(y_{i0}, \dots, y_{iT}) = (2\pi)^{-(T+1)/2} (\sigma^2)^{-(T+1)/2} |\Omega|^{-1/2} \times \exp \left\{ -\frac{1}{2\sigma^2} [y_{i0}, y_{i1} - \beta y_{i0}, \dots, y_{iT} - \beta y_{i,T-1}] \Omega^{-1} \times [y_{i0}, y_{i1} - \beta y_{i0}, \dots, y_{iT} - \beta y_{i,T-1}]' \right\}, \quad (6.1)$$

where

$$\Omega = \lambda \begin{pmatrix} \frac{1}{1-\beta^2} & \mathbf{O}' \\ \mathbf{O} & \mathbf{I}_T \end{pmatrix} + \begin{pmatrix} \frac{1}{1-\beta} \\ \mathbf{e} \end{pmatrix} \begin{pmatrix} \frac{1}{1-\beta} & \mathbf{e}' \end{pmatrix},$$

$$\Omega^{-1} = \frac{1}{\lambda} \left\{ \begin{pmatrix} 1-\beta^2 & \mathbf{O}' \\ \mathbf{O} & \mathbf{I}_T \end{pmatrix} - \left(\lambda + T + \frac{1+\beta}{1-\beta} \right)^{-1} \times \begin{pmatrix} 1+\beta \\ \mathbf{e} \end{pmatrix} (1+\beta, \mathbf{e}') \right\},$$

$$|\Omega| = \frac{\lambda^T}{1-\beta^2} \left[\lambda + T + \frac{1+\beta}{1-\beta} \right]. \quad (6.2)$$

When N is fixed and $T \rightarrow \infty$, the consistency of the MLE of β follows from the usual argument that the asymptotic theory is the same for both y_{i0} fixed and y_{i0} random with a stationary distribution (e.g., see Anderson 1971, Ch. 5). The consistency of the MLE when $N \rightarrow \infty$ and T is fixed follows from the fact that the MLE is obtained by maximizing the joint density of N independently normally distributed random vectors $(y_{i0}, y_{i1}, \dots, y_{iT})$. Again, there is no incidental parameters problem.

There are similar conditions for the occurrence of boundary solutions. The boundary solution of $\lambda = 0$ (or $\sigma_u^2 = 0$) cannot occur, but the boundary solution that $\sigma^2 = 0$ will occur if $(\mathbf{e}'\Phi^{-1}\mathbf{e}) \sum_{i=1}^N \mathbf{y}'_i \Phi^{-1} \mathbf{y}_i \geq T \cdot \sum_{i=1}^N (\mathbf{y}'_i \Phi^{-1} \mathbf{e})^2$, where the s, t th element of Φ is $\phi_{st} = \beta^{|s-t|}$ (Lee 1979).

In this case the CV will have the same consistency properties as in the other cases. As $T \rightarrow \infty$, the last term in (3.3) converges in probability to zero. As $N \rightarrow \infty$, the numerator of the last term in (3.3) converges in probability to the negative of (3.5), which is not zero. Thus the CV is consistent as $T \rightarrow \infty$ and is inconsistent as $N \rightarrow \infty$.

Although the consistency of the CV is not independent of the way T or N goes to infinity, the MLE of β in this case is consistent and independent of the way T or N goes to infinity. However, because the partial derivative equations do not have a simple solution, a complicated iterative scheme will have to be used. Unfortunately, the

assumption that w_{i0} are random with arbitrary variance $(\lambda/\eta)\sigma^2$ (subcase (b)) does not simplify the computation either. One can see this by noting that the joint density $f_i(y_{i0}, \dots, y_{iT})$ is still of the form (6.1) with the covariance matrix Ω differing from (6.2) only in the upper left element where the term $1/(1-\beta^2)$ is replaced by $1/\eta$. Thus, the MLE of β is still consistent if either $T \rightarrow \infty$ and N is fixed or $N \rightarrow \infty$ and T is fixed, but the computation is again complicated.

7. PSEUDO AND CONDITIONAL MAXIMUM LIKELIHOOD ESTIMATORS

The purpose of using a covariance estimator is to eliminate the individual effect α_i . This can be done by pre-multiplying $\mathbf{v}_i$ by the $(T-1) \times T$ transformation matrix

$$\mathbf{D}_{(T-1) \times T} = \begin{pmatrix} -1 & 1 & 0 & \dots & 0 & 0 \\ 0 & -1 & 1 & \dots & . & . \\ . & . & . & & . & . \\ . & . & . & & . & . \\ 0 & 0 & 0 & \dots & -1 & 1 \end{pmatrix}. \quad (7.1)$$

Then $\mathbf{D}\mathbf{v}_i = \mathbf{D}\mathbf{u}_i$ is normally distributed with mean $\mathbf{O}$ and covariance matrix $\sigma^2 \mathbf{D}\mathbf{A}\mathbf{D}' = \sigma_u^2 \mathbf{D}\mathbf{D}'$.

From (2.4) we have

$$y_{it} - y_{i,t-1} = \beta(y_{i,t-1} - y_{i,t-2}) + u_{it} - u_{i,t-1}. \quad (7.2)$$

From this, however, we cannot obtain MLE's as claimed by some people. This can be seen by noting that although $\mathbf{D}\mathbf{u}_i$ has a properly defined density function, $\mathbf{D}\mathbf{y}_i$ does not. Even under the assumption that y_{i0} are fixed, y_{i1} are still random. Thus, $\mathbf{D}\mathbf{y}_i$ leaves the density of $(y_{i1} - y_{i0})$ undefined.

Substituting (7.2) into the density of $\mathbf{D}\mathbf{u}_1, \dots, \mathbf{D}\mathbf{u}_N$ and maximizing this quantity with respect to β , λ , and σ^2 yields estimators that are not consistent when N tends to infinity and T is fixed. We show the inconsistency of these pseudo MLE's by considering the case in which y_{i0} are fixed and $T = 3$. Then

$$\mathbf{D}_{2 \times 3} \mathbf{u}_i = \begin{pmatrix} u_{i2} - u_{i1} \\ u_{i3} - u_{i2} \end{pmatrix} \quad (7.3)$$

is normally distributed with mean zero and variance-covariance matrix

$$\sigma^2 \mathbf{D}\mathbf{A}\mathbf{D}' = \sigma_u^2 \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \quad (7.4)$$

The determinant of $\mathbf{D}\mathbf{D}'$ is now equal to three and the inverse of it is

$$\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

If we consider (7.2), $t = 2, 3$, as a transformation from $(u_{i2} - u_{i1})$ and $(u_{i3} - u_{i2})$ to y_{i2} and y_{i3} , $i = 1, \dots, N$, the Jacobian of the transformation is one. Thus, the logarithm of the pseudo likelihood function is

$$\begin{aligned} \log \tilde{L} = \text{constant} - N \log \sigma_u^2 \\ - \frac{1}{3\sigma_u^2} \sum_{i=1}^N \{[(y_{i2} - y_{i1}) - \beta(y_{i1} - y_{i0})]^2 \\ + [(y_{i3} - y_{i2}) - \beta(y_{i2} - y_{i1})]^2 \\ + [(y_{i2} - y_{i1}) - \beta(y_{i1} - y_{i0})] \\ \times [(y_{i3} - y_{i2}) - \beta(y_{i2} - y_{i1})]\}. \end{aligned} \quad (7.5)$$

Taking partial derivatives of (7.5) with respect to β and solving for β , we obtain the pseudo MLE. The probability limit of this pseudo MLE $\hat{\beta}_{\text{PML}}$ is

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}_{\text{PML}} = \beta - \frac{2 - \beta}{\text{constant}}. \quad (7.6)$$

Hence, the estimator is inconsistent.

On the other hand, the maximization of the likelihood function obtained from the density of $(y_{i2}, \dots, y_{iT})$ conditional on y_{i1} over i does yield a consistent estimator if the unconditional MLE is consistent. This follows from the fact that conditional on y_{i1} (and y_{i0} fixed) we are maximizing the likelihood function, based on $(T - 1)$ -component independently distributed random vectors, the i th having density

$$\begin{aligned} f_i(y_{i2}, \dots, y_{iT} | y_{i1}) = (2\pi)^{-(T-1)/2} (\sigma^2)^{-(T-1)/2} |\Lambda^*|^{-1/2} \\ \times \exp\{-(1/2\sigma^2)[y_{i2} - \beta y_{i1} - [1/(1 + \lambda)](y_{i1} - \beta y_{i0}), \\ \dots, y_{iT} - \beta y_{i,T-1} - [1/(1 + \lambda)](y_{i1} - \beta y_{i0})\Lambda^{*-1} \\ [y_{i2} - \beta y_{i1} - [1/(1 + \lambda)](y_{i1} - \beta y_{i0}), \\ \dots, y_{iT} - \beta y_{i,T-1} - [1/(1 + \lambda)](y_{i1} - \beta y_{i0})]'], \\ i = 1, \dots, N, \end{aligned} \quad (7.7)$$

where

$$\begin{aligned} \Lambda^* = (\lambda I_{T-1} + \mathbf{e}_{T-1} \mathbf{e}'_{T-1}) \\ - [1/(1 + \lambda)] \mathbf{e}_{T-1} \mathbf{e}'_{T-1}. \end{aligned} \quad (7.8)$$

The conditional MLE is consistent when T is fixed and $N \rightarrow \infty$. It is the pseudo MLE that is inconsistent. We suspect it is this confusion about the proper form of the conditional likelihood function that caused the confusion about the consistency of the conditional MLE (e.g., Chamberlain 1979; Lee 1979).

8. SIMPLE CONSISTENT ESTIMATES

Although the maximization of the density of $\mathbf{Dy}_1, \dots, \mathbf{Dy}_N$ does not yield consistent estimates, it does suggest some simple consistent estimators. From (7.2) we know that we may use either $y_{i,t-2}$ or $(y_{i,t-2} - y_{i,t-3})$

as instruments and estimate β by

$$\hat{\beta}_{\text{IV}} = \frac{\sum_{i=1}^N \sum_{t=3}^T (y_{it} - y_{i,t-1})(y_{i,t-2} - y_{i,t-3})}{\sum_{i=1}^N \sum_{t=3}^T (y_{i,t-1} - y_{i,t-2})(y_{i,t-2} - y_{i,t-3})} \quad (8.1)$$

or

$$\tilde{\beta}_{\text{IV}} = \frac{\sum_{i=1}^N \sum_{t=2}^T (y_{it} - y_{i,t-1})y_{i,t-2}}{\sum_{i=1}^N \sum_{t=2}^T (y_{i,t-1} - y_{i,t-2})y_{i,t-2}}. \quad (8.2)$$

Both (8.1) and (8.2) are consistent when $N \rightarrow \infty$ or $T \rightarrow \infty$ or both.

Estimator (8.2) has the advantage over (8.1) in the sense that the minimum time period required is two, while (8.1) requires $T \geq 3$. However, (8.1) and (8.2) have different asymptotic variances. Under the assumption that y_{i0} are random with a stationary distribution (Sec. 6), the asymptotic variance of (8.1) is

$$\begin{aligned} \text{asy var} [\sqrt{N}(\hat{\beta}_{\text{IV}} - \beta)] \\ = \frac{4}{T-2} \cdot \frac{(1 - \beta^2)(1 - \beta)}{(1 + \beta^2)^2}, \end{aligned} \quad (8.3)$$

the asymptotic variance of (8.2) is

$$\begin{aligned} \text{asy var} [\sqrt{N}(\tilde{\beta}_{\text{IV}} - \beta)] \\ = \frac{2(1 + \beta)}{T-1} \left[\frac{1}{1 - \beta} + \frac{1 + \beta}{(1 - \beta)^2 \lambda} \right]. \end{aligned} \quad (8.4)$$

Therefore, (8.1) is preferable to (8.2) if

$$\begin{aligned} 2(T-1)/(T-2) < [(1 + \beta^2)^2/(1 - \beta)^4] \\ \times [(1 - \beta) + (1 + \beta)/\lambda]. \end{aligned} \quad (8.5)$$

Without knowledge of β and λ , there is little to choose between these two estimators. It appears, however, that (8.5) is more likely to be satisfied if β is positive. Thus, as a rough rule of thumb we may want to use (8.1) if there is prior belief that successive observations are positively correlated and use (8.2) if successive observations are negatively correlated.

Once a consistent estimator of β is obtained, we may substitute it into (2.4) to estimate λ and σ^2 . Substituting the estimated λ into Λ in (3.7), we can reestimate β by the generalized least squares method. This two-round estimator is similar to the one suggested by Nerlove (1971), except in the first round we use a different and simpler instrument. It is also equivalent to the MLE of β under the assumption that λ are known and y_{i0} are fixed.

As we have seen, different assumptions about the initial observations do not affect the consistency of the MLE's when $T \rightarrow \infty$. However, a typical panel usually involves a large number of individuals, but only over a short period of time. As it turns out, the properties of the MLE depend

Table 1. Interpretations and Statistical Properties of the MLE's and CV's for Models Under Different Assumptions About the Initial Observations^a

Interpretation of the model	Case I: y_{i0} Fixed		Case II: y_{i0} Random With a Common Mean		Case III: w_{i0} Fixed		Case IV: w_{i0} Random With a Stationary Distribution	
	$T \rightarrow \infty$ N fixed	T fixed $N \rightarrow \infty$	$T \rightarrow \infty$ N fixed	T fixed $N \rightarrow \infty$	$T \rightarrow \infty$ N fixed	T fixed $N \rightarrow \infty$	$T \rightarrow \infty$ N fixed	T fixed $N \rightarrow \infty$
Statistical properties of	MLE	Consistent	Consistent	Consistent	Consistent	Inconsistent	Consistent	Consistent
	CV	Consistent	Inconsistent	Consistent	Inconsistent	Inconsistent	Consistent	Inconsistent

^a In Case III the MLE should be interpreted as the interior solution.

crucially on the assumption of the initial conditions. Different assumptions about the initial conditions call for different methods to obtain the MLE. Mistaking one case for the other will generally not lead to asymptotically equivalent formulas. Consequently, the misused estimator may be inconsistent. Unfortunately, we usually have little information to rely on in making a correct choice of the initial conditions. Estimator (8.1) or (8.2) has the advantage that it is consistent independent of what the initial conditions are. Thus, the instrumental variable method, although is less efficient, does have merit. Furthermore, if we know the correct choice of the initial conditions, we can always use the instrumental variable estimates as the initial value to start the iterative process to obtain the more efficient MLE.

9. CONCLUSIONS

In this paper we have studied the problems of estimating a dynamic model with error components in panel data when either the number of time point T or the number of cross-sectional unit N tends to infinity. We examined several models arising from different assumptions about the initial conditions. We attempted an interpretation and studied the properties of the MLE's and covariance estimators for each of these models. The main conclusions are summarized in Table 1. The confusion between the pseudo and conditional MLE in the literature was also clarified in Section 7.

As we can see from Table 1, the MLE is consistent when T tends to infinity no matter what the assumptions about the initial conditions are. When T is fixed and N tends to infinity, the consistency of the MLE will depend on the assumptions about the initial conditions.

On the other hand, the covariance estimator does not depend on the initial conditions. When $T \rightarrow \infty$, it is always consistent. When T is fixed, it is always inconsistent, no

matter what our initial conditions are. Because the justification of using the covariance estimator for a dynamic model mainly rests on the asymptotic properties as the length of series $T \rightarrow \infty$ and the typical panel has a large number of individuals observed over a short period, it appears that the case for using the covariance estimator is not favorable.⁸

A favorable aspect of the MLE is its desirable asymptotic properties (when T or N or both tend to infinity) in most circumstances; however, the computations of the MLE's are complicated. In the special case in which the individual effects may be viewed as the effect of the initial observation or initial endowment (corrected for the mean) and may affect the individual equilibrium level (Sec. 4), the conditional MLE becomes very simple. We only need to modify the dependent variable as the actual subtracting the initial observation and apply the least squares regression to the transformed model.

It should be noted that the method of obtaining the MLE is different under different assumptions about the initial conditions. Mistaking one case for the other will not give us a consistent estimator as $N \rightarrow \infty$. A simple instrumental variable method was therefore suggested in Section 8. Although it is less efficient, it does have the advantage that it is consistent independent of the initial conditions.

In this paper we have concentrated on the asymptotic properties of different estimators. There is no guarantee that in small samples the estimators will possess finite second-order moments. Further study on this problem is needed. Some modifications of these estimators along the lines suggested by Fuller (1977), Mehta and Swamy (1978), and Swamy and Rappoport (1978), guaranteeing

⁸ Swamy and Mehta (1979) in a different setup have also shown that the CV is not necessarily a desirable estimator under certain conditions.

the existence of moments in small samples, are highly desirable.

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